Experiment -3

Python Program To Solve Water Jug Problem

Code:

```
from collections import deque

def water_jug(a, b, target):
    queue = deque([(0, 0)])
    visited = set()

    while queue:
        x, y = queue.popleft()

        if (x, y) in visited:
            continue

        visited.add((x, y))
        print(x, y)

        if x == target or y == target:
            print("Goal reached!")
            return

        states = [
            (a, y),
            (x, b),
            (0, y),
            (x, 0),
```

```
            (x - min(x, b-y), y + min(x, b-y)),
            (x + min(y, a-x), y - min(y, a-x))
        ]

        for state in states:
            if state not in visited:
                queue.append(state)

    print("No solution")

water_jug(4, 3, 2)
```

```python
from collections import deque

def water_jug(a, b, target):
    queue = deque([(0, 0)])
    visited = set()

    while queue:
        x, y = queue.popleft()

        if (x, y) in visited:
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        states = [
            (a, y),
            (x, b),
            (0, y),
            (x, 0),
            (x - min(x, b-y), y + min(x, b-y)),
            (x + min(y, a-x), y - min(y, a-x))
        ]

        for state in states:
            if state not in visited:
                queue.append(state)

    print("No solution")

water_jug(4, 3, 2)
```

```
0 0
4 0
0 3
4 3
1 3
3 0
1 0
3 3
0 1
4 2
Goal reached!


...Program finished with exit code 0
Press ENTER to exit console.
```